

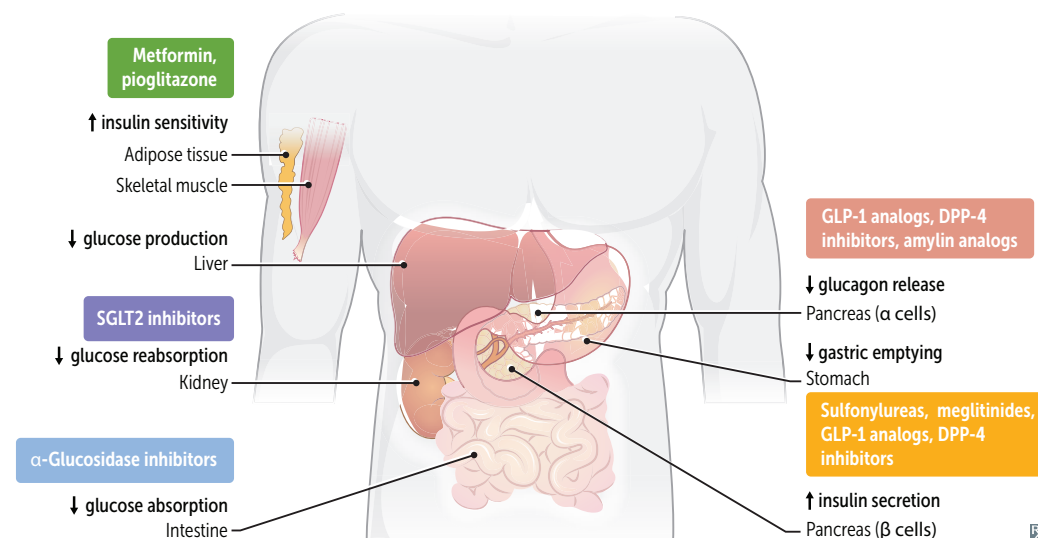
► ENDOCRINE—PHARMACOLOGY

Diabetes mellitus therapy

All patients with diabetes mellitus should receive education on diet, exercise, blood glucose monitoring, and complication management. Treatment differs based on the type of diabetes and glycemic control:

- Type 1 DM—insulin replacement
- Type 2 DM—oral agents (metformin is first line), non-insulin injectables, insulin replacement; weight loss particularly helpful in lowering blood glucose
- Gestational DM—insulin replacement if nutrition therapy and exercise alone fail

Regular (short-acting) insulin is preferred for DKA (via IV), hyperkalemia (with glucose to prevent concomitant hypoglycemia), and stress hyperglycemia.



DRUG	MECHANISM	ADVERSE EFFECTS
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Insulin preparations**Rapid acting (no lag):**

lispro, aspart, glulisine

Short acting:

regular

Intermediate acting:

NPH

Long acting:

detemir, glargine

Very long acting:

degludec

Bind insulin receptor (tyrosine kinase activity).

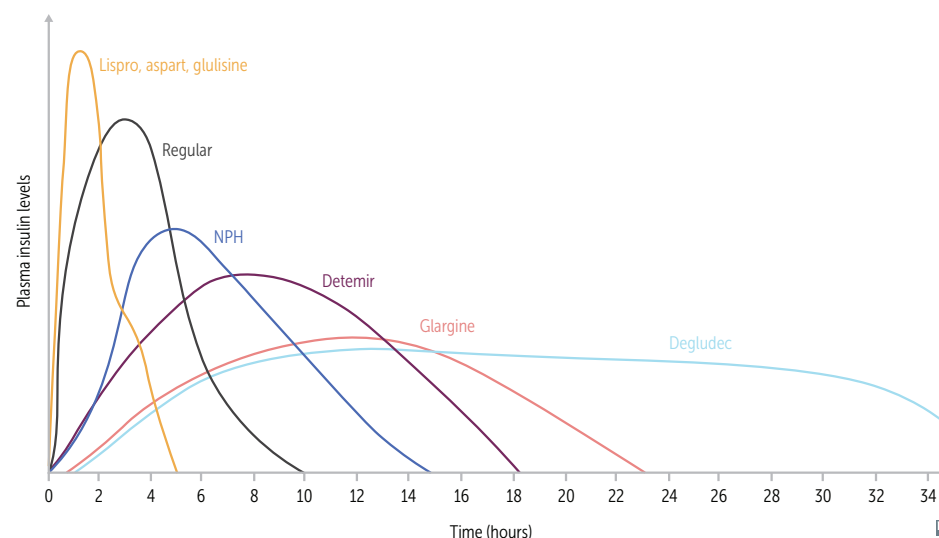
Liver: ↑ glucose storage as glycogen.

Muscle: ↑ glycogen, protein synthesis.

Fat: ↑ TG storage.

Cell membrane: ↑ K⁺ uptake.

Hypoglycemia, lipodystrophy, hypersensitivity reactions (rare), weight gain.



Diabetes mellitus therapy (continued)

DRUG	MECHANISM	ADVERSE EFFECTS
Increase insulin sensitivity		
Metformin	Inhibits mitochondrial glycerol-3-phosphate dehydrogenase (mGPD) → inhibition of hepatic gluconeogenesis and the action of glucagon. ↑ glycolysis, peripheral glucose uptake (↑ insulin sensitivity).	GI upset, lactic acidosis (in patients with renal insufficiency, cirrhosis, sepsis, or hypoperfusion), vitamin B ₁₂ deficiency. Weight loss (often desired).
Pioglitazone	Activate PPAR-γ (a nuclear receptor) → ↑ insulin sensitivity and levels of adiponectin → regulation of glucose metabolism and fatty acid storage.	Weight gain, edema, HF, ↑ risk of fractures. Delayed onset of action (several weeks).
Increase insulin secretion		
Sulfonylureas (1st gen) Chlorpropamide, tolbutamide	Close K ⁺ channels in pancreatic B cell membrane → cell depolarizes → insulin release via ↑ Ca ²⁺ influx.	Disulfiram-like reaction with first-generation sulfonylureas only (rarely used). Hypoglycemia (↑ risk in renal insufficiency), weight gain.
Sulfonylureas (2nd gen) Glipizide, glyburide		
Meglitinides Nateglinide, repaglinide		
Increase glucose-induced insulin secretion		
GLP-1 analogs Exenatide, liraglutide, semaglutide	↓ glucagon release, ↓ gastric emptying, ↑ glucose-dependent insulin release.	Nausea, vomiting, pancreatitis. Weight loss (often desired). ↑ satiety (often desired).
DPP-4 inhibitors Linagliptin, saxagliptin, sitagliptin	Inhibit DPP-4 enzyme that deactivates GLP-1 → ↓ glucagon release, ↓ gastric emptying. ↑ glucose-dependent insulin release.	Respiratory and urinary infections, weight neutral. ↑ satiety (often desired).
Decrease glucose absorption		
Sodium-glucose co-transporter 2 inhibitors Canagliflozin, dapagliflozin, empagliflozin	Block reabsorption of glucose in proximal convoluted tubule (glucosuria). Also cause natriuresis, osmotic diuresis → ↓ blood glucose, ↓ serum Na ⁺ , ↓ ECF volume, ↓ blood pressure.	Glucosuria (UTIs, vulvovaginal candidiasis), dehydration (orthostatic hypotension), weight loss. Glucose flows in urine. Use with caution in renal insufficiency (↓ efficacy with ↓ GFR).
α-glucosidase inhibitors Acarbose, miglitol	Inhibit intestinal brush-border α-glucosidases → delayed carbohydrate hydrolysis and glucose absorption → ↓ postprandial hyperglycemia.	GI upset, bloating. Not recommended in renal insufficiency.
Others		
Amylin analogs Pramlintide	↓ glucagon release, ↓ gastric emptying.	Hypoglycemia, nausea. ↑ satiety (often desired).

Thionamides

Propylthiouracil, methimazole.

MECHANISM	Block thyroid peroxidase, inhibiting the oxidation of iodide as well as the organification and coupling of iodine → inhibition of thyroid hormone synthesis. PTU also blocks 5'-deiodinase → ↓ P eripheral conversion of T ₄ to T ₃ .
CLINICAL USE	Hyperthyroidism. PTU used in P rimary (first) trimester of pregnancy (due to methimazole teratogenicity); methimazole used in second and third trimesters of pregnancy (due to risk of PTU-induced hepatotoxicity). Not used to treat Graves ophthalmopathy (treated with glucocorticoids).
ADVERSE EFFECTS	Skin rash, agranulocytosis (rare), aplastic anemia, hepatotoxicity. PTU use has been associated with ANCA-positive vasculitis. Methimazole is a possible teratogen (can cause aplasia cutis).

Levothyroxine, liothyronine

MECHANISM	Hormone replacement for T ₄ (levo thyroxine; levo = 4 letters) or T ₃ (lio thyronine; lio = 3 letters). Avoid levothyroxine with antacids, bile acid resins, or ferrous sulfate (↓ absorption).
CLINICAL USE	Hypothyroidism, myxedema. May be misused for weight loss. Distinguish exogenous hyperthyroidism from endogenous hyperthyroidism by using a combination of TSH receptor antibodies, radioactive iodine uptake, and/or measurement of thyroid blood flow on ultrasound.
ADVERSE EFFECTS	Tachycardia, heat intolerance, tremors, arrhythmias.

Hypothalamic/pituitary drugs

DRUG	CLINICAL USE
Conivaptan, tolvaptan	ADH antagonists SIADH (block action of ADH at V ₂ -receptor)
Demeclocycline	ADH antagonist, a tetracycline SIADH (interferes with ADH signaling)
Desmopressin	ADH analog Central DI, von Willebrand disease, sleep enuresis, hemophilia A
GH	GH deficiency, Turner syndrome
Oxytocin	Induction of labor (stimulates uterine contractions), control uterine hemorrhage
Octreotide	Somatostatin analog Acromegaly, carcinoid syndrome, gastrinoma, glucagonoma, esophageal varices

Fludrocortisone

MECHANISM	Synthetic analog of aldosterone with glucocorticoid effects. Fluid rocortisone retains fluid .
CLINICAL USE	Mineralocorticoid replacement in 1° adrenal insufficiency.
ADVERSE EFFECTS	Similar to glucocorticoids; also edema, exacerbation of heart failure, hyperpigmentation.

Cinacalcet

MECHANISM	Sensitizes calcium -sensing receptor (CaSR) in parathyroid gland to circulating $\text{Ca}^{2+} \rightarrow \downarrow \text{PTH}$. Pronounce “ Senacalcet .”
CLINICAL USE	2° hyperparathyroidism in patients with CKD receiving hemodialysis, hypercalcemia in 1° hyperparathyroidism (if parathyroidectomy fails), or in parathyroid carcinoma.
ADVERSE EFFECTS	Hypocalcemia.

Sevelamer

MECHANISM	Nonabsorbable phosphate binder that prevents phosphate absorption from the GI tract.
CLINICAL USE	Hyperphosphatemia in CKD.
ADVERSE EFFECTS	Hypophosphatemia, GI upset.

Cation exchange resins Patiromer, sodium polystyrene sulfonate, zirconium cyclosilicate.

MECHANISM	Bind K^+ in colon in exchange for other cations (eg, Na^+ , Ca^{2+}) $\rightarrow \text{K}^+$ excreted in feces.
CLINICAL USE	Hyperkalemia.
ADVERSE EFFECTS	Hypokalemia, GI upset.